Geometric Torsion Beyond Curves

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Geometric torsion over a higher-dimensional base

Let Z be a quasi-projective variety of dimension m, let F be totally real with $[F : \mathbb{Q}] = n$, and let $A \to Z$ be a non-isotrivial family of n-dimensional abelian varieties with real multiplication by $\mathcal{O}_F$.

If $\overline{Z}$ is any smooth projective model of Z, its canonical volume is

$$\widetilde{\mathrm{vol}}(Z) := \mathrm{vol}(K_{\overline{Z}}) = \limsup_{b\to\infty} \frac{h^0(\overline{Z}, bK_{\overline{Z}})}{b^m/m!}.$$

For an integral curve of geometric genus g, $\widetilde{\mathrm{vol}}(Z) = \max\{2g-2, 0\}$.

theorem (M.)

For every $v \geq 0$ and $n \geq 1$, there exists $N = N(v, n)$ such that

$$\widetilde{\mathrm{vol}}(Z) \leq v \qquad \Longrightarrow \qquad A(Z)_{\mathrm{tors}} \subseteq A(Z)[N].$$

When Z is a curve, this is proved by Bakker and Tsimerman in 2015.

Large level forces every subvariety to be of general type

Let $X(1)$ be the Hilbert modular variety. For a torsion point P, set $\eta = \mathrm{Ann}_{\mathcal{O}_F}(P)$. The level cover $X_1(\eta) \to X(1)$ parametrizes abelian varieties with real multiplication together with such a point P.

corollary (M.)

Fix n. For every totally real field $F/\mathbb{Q}$ of degree n, every positive-dimensional subvariety of $X_1(\eta)$ is of general type provided that

$$|\,\mathrm{Nm}(\eta)| > (2\pi)^{2n}.$$

Bombieri–Lang conjecture predicts that, for every number field K and every such ideal η,

$$\#X_1(\eta)(K) < \infty.$$

Unconditionally, rational points become sparse

Let $L = K_{\overline{X}_1(\eta)} + D$ on a smooth toroidal compactification.

corollary (M.)

Suppose $X_1(\eta)$ does not have an elliptic point. Let $\epsilon > 0$ and $B \geq \epsilon[K : \mathbb{Q}]$. Then there exists $c = c(K, \epsilon) > 0$ such that

$$\#\Big\{x \in X_1(\eta)(K) : H_L(x) \leq B\Big\} \leq cB^{\delta},$$

where

$$\delta = \frac{(n+1)[K : \mathbb{Q}]}{4\sinh^2\big(\frac{1}{n}\log|\operatorname{Nm}(\eta)| - 1\big)}(1+\epsilon).$$

For fixed n and $[K : \mathbb{Q}]$, $\delta \to 0$ as $|\operatorname{Nm}(\eta)| \to \infty$.